

Lesson 7: Reading I-V graphs

Electric Circuits · Year 10 · 50 minutes · AQA 6.2.1.4

Why this lesson is here

Three components share the same axes but produce very different curves. Their shape tells you how their resistance behaves.

Learning objective

Recognise characteristic graph shapes and explain what they show.

Core teaching point

On I (vertical) against V (horizontal): fixed resistor is a straight line through origin at constant temperature; lamp curve flattens as it heats; diode conducts mainly one way.

Equipment and preparation

Printed/sketched I-V axes, graph paper, optional lamp demonstration; existing I-V practical worksheet.

Lesson sequence

Time	Phase	Teacher move and evidence
0-6	Retrieval	Calculate R at a graph point.
6-19	Teach shapes	Sketch resistor, lamp and diode with axes specified.
19-31	Compare	Students sort unlabelled curves and justify each.
31-43	Practice	Tasks 2-4, including two-point calculation.
43-50	Check	Four-mark lamp question.

Workbook task guidance

Task 1: On a blank set of axes, sketch and label the three characteristic shapes in both directions.

Expected: Resistor straight line through origin; lamp symmetric curve flattening at high |V|; diode near zero reverse current and steep forward rise.

Task 2: Explain why a filament lamp graph becomes less steep as p.d. rises.

Expected: Greater current heats the filament; resistance rises, so extra V gives proportionally less current.

Task 3: At 2.0 V a component carries 0.10 A; at 4.0 V it carries 0.15 A. Calculate R at each point and decide whether it is ohmic.

Expected: $R_1 = 20 \text{ ohm}$; $R_2 = 26.7 \text{ ohm}$, so not constant R.

Task 4: Describe what reversing a diode does to its current.

Expected: Reverse current is very small compared with forward current (ignoring breakdown).

Worked example

A graph point is $V = 4.0 \text{ V}$, $I = 0.20 \text{ A}$. At that point $R = 4.0/0.20 = 20 \text{ ohm}$. Do not infer R is constant from one point.

AQA-style checkpoint

[4 marks] A lamp carries 0.20 A at 2.0 V and 0.30 A at 6.0 V. Calculate resistance at each point and explain the change.

Mark guidance: 10 ohm (1); 20 ohm (1). Larger current heats filament (1), raising resistance (1).

Anticipated misconception

For I on the y-axis, a flatter graph indicates a larger resistance; always check the axes.

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.